Tanmay Bhuskute

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Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building LLM applications, RAG pipelines, backend APIs, and cloud automation with Python, AWS, and Kubernetes. Researching LLM-powered Android testing.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Machine Learning, Data Mining, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL, Go, C/C++.

Web & Backend: REST APIs, FastAPI, Flask, Node.js, Next.js, React, Django, HTML/CSS.

Databases: PostgreSQL, MySQL, Supabase, Redis, Vector Databases.

Developer Tools: Git, GitHub, Docker, GitLab CI/CD, AWS, Kubernetes, Terraform, Azure, Google Cloud Platform.

AI & Data: LLMs, RAG, LangChain, Prompt Engineering, MCP, Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, pandas.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented **LLM agent pipeline** on Gemini 2.5 Pro and Vertex AI, decoupling video understanding from closed-loop Android device automation, evaluated against **ViBR** across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Developed **Go + Terratest** integration tests, identifying about **95%** of historical misconfiguration scenarios.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built **React.js and MySQL** records app with on-premises database, cutting retrieval time **90% (5 min to 30 sec)**.

PROJECTS AND RESEARCH

PolicyPilot | Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

Spendora | AI Personal Finance Platform [Next.js, TS, Supabase, Gemini] Jun 2025

- Engineered **Gemini-powered, schema-validated** transaction workflows for natural-language, voice, and receipt input, cutting median expense-entry time **72% (29s to 8s)**.
- Benchmarked **LLM financial assistant** on 100 queries: **94% numerical correctness**, **90% tool-selection accuracy**.
- Implemented Supabase auth, row-level security, and AI usage quotas, backed by **80 automated tests**.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).